Speaker Notes:
Hospital Chatbot for Color-Coded Clinical Pathways

Core explanation of every presentation slide

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Abstract

These notes walk through the final Beamer deck slide by slide. For each topic you get: what the slide means in plain language, why it is in the presentation, what to say aloud, and—where needed—the mathematics explained from the ground up. Use this document to prepare for the oral presentation; it is not a second thesis.

Contents

1 Title, outline, and background

1.1 Title page and outline (slides 1–2)

Why this slide

The title page names the project, authors, supervisor, and department. The outline shows the five big blocks of the talk: Background, Problem/Objectives, Methodology (including mathematics), Results, Impact.

What to say

"Today we present a hospital chatbot that reads an English chief complaint and returns a SATS colour plus a pathway card for KNUST University Hospital. The talk moves from the hospital problem, through the mathematics of BioBERT triage, to holdout results and anticipated impact."

Core idea of the project. Patients often arrive with *words only*. Nurses later measure vitals (TEWS). Our system acts at the text-first moment: free-text complaint $X \to$ acuity probabilities $\hat{\mathbf{y}} \to$ colour $C_{\mathrm{NLP}} \to$ pathway card $P(C)$. Nurses keep TEWS and the right to override.

1.2 Hospital overcrowding photo (slide 3)

Why this slide

A visual of overcrowded wards grounds the problem in Ghanaian hospital reality before any equations. Source attribution: GhanaFact fact-check article on Korle-Bu overcrowding imagery.

What to say

"This is the world our system sits in: too many patients, limited beds and staff. Triage has to decide who is seen first and where they go next."

1.3 Hospital intake under pressure (slide 4)

Why this slide

Defines the operational setting: crowded queues, SATS at teaching hospitals including KNUST University Hospital, and *acuity discontinuity*—colour assigned at the first desk may not travel with the patient.

Acuity discontinuity means: urgency is decided once at intake, then often forgotten when the patient moves to imaging, labs, or wards. Colour and countdown do not reliably steer later decisions (Rominski et al., 2014; KATH Emergency Department, 2018).

What to say

"Even when SATS is used, urgency can be lost after the first desk. That is the discontinuity we want a digital colour-plus-pathway record to reduce."

1.4 SATS colour bands (slide 5)

Why this slide

Everyone in the room must share the same colour language before you talk about NLP.

Definition 1 (SATS colours)**.** *The South African Triage Scale uses five colours (South African Triage Group, 2012):*

- ***Red:*** *immediate / resuscitation stream.*
- ***Orange:*** *very urgent (about < 10 minutes).*
- ***Yellow:*** *urgent (about < 60 minutes).*
- ***Green:*** *routine / outpatient stream.*
- ***Blue:*** *deceased-on-arrival dignity protocol (not "low urgency").*

Our chatbot maps predicted acuity levels to Red/Orange/Yellow/Green for living patients. Blue is a dignity protocol, not an NLP output class in this prototype.

1.5 Text first, then vitals (slide 6)

Why this slide

Separates the two clocks of triage: words arrive before thermometers.

- **Chief complaint:** primary reason for visit, in the patient's (or attendant's) own words.
- **TEWS:** nurse score from HR, RR, mobility, temperature, AVPU, trauma → colour from vitals.
- **This project:** English text → colour + pathway *before* vitals; TEWS stays in parallel.

What to say

"We do not replace the nurse. We give a first acuity signal from English text while TEWS remains the vital-sign standard."

1.6 Literature review (slide 7)

Why this slide

Shows you know the field: overcrowding/SATS, TEWS limits, BioBERT lineage, NLP triage gaps.

Four points to master.

1. LMIC overcrowding and SATS with waiting times $T_{\max}$ (KATH Emergency Department, 2018; South African Triage Group, 2012); text-first arrival still under-formalised.
2. TEWS is objective and does **not** read language (South African Triage Group, 2012).
3. Self-attention → BERT → BioBERT on PubMed/PMC (Vaswani et al., 2017; Lee et al., 2020).
4. NLP at ED triage is active but mostly high-income; local SATS+pathway math is thin (Stewart et al., 2023; Gligorijević et al., 2018).

What to say

"The literature gives us BioBERT and triage NLP abroad. What is missing is an examinable map from Ghanaian intake text to SATS colour and a local pathway card—that is our gap."

2 Problem statement, research gap, and objectives

2.1 Problem statement (slide 8)

Why this slide

States the pain in two blocks: lost urgency for Red/Orange patients, and Green congestion blocking capacity.

Lost urgency. Critical patients can lose time-critical priority when they move between imaging, labs, and wards. There is often no digital record that carries colour and a countdown timer with them.

Green congestion. Non-urgent patients lack automated guidance to outpatient or self-care routes. They occupy space and staff needed for life-saving care.

What to say

"Two failures: urgent patients lose their colour as they move, and non-urgent patients clog the stream. We want a digital colour and pathway that travels with the patient."

2.2 Research gap (slide 9)

Why this slide

This is the academic justification. Examiners listen carefully here.

The gap has three layers:

1. Formal mathematics from patient text $\rightarrow$ SATS colour $\rightarrow$ local pathway for Ghanaian intake is under-specified.
2. Most NLP triage papers use high-income corpora; local department routing is rarely formalised.
3. TEWS covers vitals; the text-first moment lacks an examinable pipeline.

What this thesis fills.

$$X \rightarrow \hat{\mathbf{y}} \rightarrow C_{\text{NLP}} \rightarrow P(C)$$

at KNUST University Hospital, for English chief complaints.

What to say

"We fill the gap with an end-to-end, examinable chain: English complaint to probabilities, colour, and pathway card."

2.3 Objectives (slide 10)

Why this slide

Maps the thesis aims to what the chatbot does.

General objective: develop a hospital chatbot that classifies unstructured English chief complaints into SATS colours and clinical pathways at intake.

Specific objectives (remember the four):

1. Automate triage classification from a single chief complaint.
2. Guide clinical pathways (urgent vs routine streams).
3. Integrate with SATS and KNUST University Hospital routing.
4. Support refinement via visit-informed pathway design.

What to say

"Our aim is one sentence: from one English chief complaint, produce a colour and a pathway that matches SATS-style routing at KNUST University Hospital."

3 Methodology overview (before the equations)

The Methodology section of the talk has two layers: (i) what data and system pieces you use, (ii) the mathematics of each pipeline stage. This chapter covers layer (i). The next chapter goes to the mathematical core.

3.1 Chief complaints as input (slide 11)

Definition 2 (Chief complaint X)**.** *The primary reason the patient presents, in their own words, as unstructured free text at intake. It is* not *a diagnosis, not TEWS, and not a full medical history.*

Why this slide

Forces clarity on the input variable. Everything later is a function of X.

Why English only? BioBERT's WordPiece vocabulary and the Triagegeist training pairs are English. The prototype does not claim Twi or mixed-language support.

Running example. At KNUST University Hospital we reuse: *"I have a headache and feel feverish. My body aches and I feel weak."* Walk this sentence through gate → tokens → colour when an examiner asks for an example.

What to say

"Our only input is the English chief complaint text X. Vitals are the nurse's job in parallel."

3.2 Training data provenance (slide 12)

Why this slide

Stops the "where did the numbers come from?" question. Two layers must never be confused.

Layer A — Triagegeist (metrics source). Synthetic ED benchmark (Laitinen-Fredriksson Foundation, 2025). Merge `train.csv` + `chief_complaints.csv` $\rightarrow N = 80{,}000$ English pairs (complaint text, acuity label). Stage 3 project fine-tuning and the 90/10 holdout use these pairs. **All reported accuracy / F1 / recall numbers come from Layer A.**

Layer B — KNUST visit examples. Written after a site visit to KNUST University Hospital (workflow, SATS colours, departments). Used for pathway demos and qualitative checks. **Not** merged into the 80k training set—so they do not inflate metrics.

What to say

"Eighty thousand Triagegeist pairs train and evaluate the classifier. KNUST examples shape pathways and demos; they are not in the training metrics."

3.3 NLP pipeline overview (slide 13)

Why this slide

One diagram for the whole story before equations. Examiners should see the chain once.

Pipeline stages in order:

$$X \rightarrow \text{clinical gate} \rightarrow \text{tokenise} \rightarrow \text{embed (768-d)} \rightarrow \text{BioBERT } (L = 12) \rightarrow \text{softmax (5 classes)} \rightarrow \text{SATS}$$

- Gate rejects non-clinical text (greetings) before BioBERT runs.
- Softmax produces five probabilities that sum to 1.
- Chosen level $\hat{c}$: the class with the *highest* probability.
- Fixed map $C_{\text{NLP}} = f_{\text{SATS}}(\hat{c})$, then pathway card $P(C_{\text{NLP}})$.

What to say

"Gate first, then BioBERT, then pick the most likely acuity level, map it to a colour, and attach a pathway card."

3.4 Pathways at KNUST University Hospital (slide 14)

Why this slide

Connects colours to destinations and timers—the clinical product, not only a class label.

Colour	Destination (summary)	Timer
Gate reject	Re-prompt for clinical text	n/a
Red	Resuscitation / immediate	Immediate
Orange	Acute / high-dependency	$\approx$ 10 min
Yellow	Urgent waiting / OPD urgent	$\approx$ 60 min
Green	Routine OPD / digital queue	Longer

Seventeen hospital units were mapped (clinical + support). Pathway cards carry colour, destination, timer, and escalation.

What to say

"A colour alone is not enough. The card says where to go and how soon."

3.5 System architecture (slide 15)

Why this slide

Shows the software path and that nurses remain in the loop.

Solid path: patient/attendant → web UI → `/predict` API → gate → BioBERT → pathway card → department routing.

Dashed path: nurse TEWS in parallel into routing. Staff can override.

What to say

"The chatbot is an assistive API in a web UI. TEWS and clinical override stay with the nurse."

4 Mathematics of the pipeline (slides 16–21)

This chapter is the mathematical core of the talk. Every slide equation is expanded from first principles: definitions, intermediate steps, a worked example on *"I have a headache and feel feverish. My body aches and I feel weak."*, and why the design exists. Non-math slides were covered earlier; here we stay inside the BioBERT triage chain.

4.1 Tokenisation (slide 16, part 1)

Intuition

Neural networks do not read English paragraphs. They read fixed-length lists of integer IDs. Tokenisation is the map from raw complaint text X to that list.

Why this slide

Without tokenisation there is no BioBERT input. This answers: how does text become numbers?

Definition 3 (Tokenisation map)**.**

$$X \xrightarrow{\mathcal{T}} (t_0, t_1, \ldots, t_{M-1}), \qquad M \leq 128.$$

Each t_i is a subword token (or a special token) with an integer ID in the vocabulary $\mathcal{V}$, $|\mathcal{V}| \approx$ 28,996.

Why $M \le 128$? BioBERT base was trained with this maximum length. Shorter complaints are padded with `<PAD>`; longer ones are truncated. Fixed length lets the hospital batch many requests through identical matrix shapes.

WordPiece algorithm. WordPiece keeps common clinical words whole ("fever", "pain") and splits rare or misspelled forms into pieces, e.g. "headache" $\rightarrow$ `head` + `##ache`. Each piece occupies its own position, so M counts *subword tokens*, not raw English words. That keeps $\mathcal{V}$ finite while still representing novel spellings at intake.

Three special tokens.

Token	ID	Purpose
`[CLS]`	101	Classification summary; always at position 0
`[SEP]`	102	End-of-sentence boundary before padding
`<PAD>`	0	Fill to length 128; attention mask zeros these positions

Worked example (running complaint). For *"I have a headache and feel feverish. My body aches and I feel weak."* (simplified tokenisation as in the thesis):

i	Raw	Token	ID
0	n/a	`[CLS]`	101
1	I	i	1045
2	have	have	2031
3	a	a	1037
4–5	headache	head + `##ache`	7994, 8772
6	and	and	1998
7	feel	feel	2514
8–9	feverish	fever + `##ish`	9643, 6804
10	.	.	1012
11	n/a	`[SEP]`	102
12–127	n/a	`<PAD>`	0

Notice: "headache" and "feverish" each cost *two* positions. That is why we say M counts subwords.

What to say

"We split the English complaint with WordPiece, insert `[CLS]` and `[SEP]`, and pad to 128 IDs. That list is the only thing BioBERT can read."

4.2 The `[CLS]` summary vector (slide 16, part 2)

Intuition

Sequence classification needs one fixed-length vector for the whole complaint. Averaging every token would weight "I" and "and" the same as "feverish". BERT instead uses an artificial summary token `[CLS]` at position 0.

After $L = 12$ encoder layers, every token row has been updated by attending to every other token. The row at position 0 is

$$h_{[\mathrm{CLS}]} = H^{(L)}_{0,:} \in \mathbb{R}^{768}. \tag{1}$$

The five-class head reads *only* this vector. `[SEP]` marks where real text ends so padding is not treated as symptoms; `[SEP]` itself is not used for acuity prediction.

Remark 1. *Self-attention updates the [CLS] row at every layer, so by layer 12 it encodes information from all symptom tokens. That is why one 768-vector can represent the full clinical wording of X.*

What to say

"After twelve layers we take the first row—`[CLS]`—as a 768-number summary of the whole sentence."

4.3 Embedding lookup and the three-sum (slide 17)

Intuition

A token ID is only an index into a dictionary of meaning vectors. We also tell the model *where* the token sits and which segment it belongs to.

Why this slide

Explains how discrete IDs become continuous vectors before attention starts.

Word embedding. The matrix $E_{\text{word}} \in \mathbb{R}^{V\times 768}$ maps each ID to a dense vector:

$$\mathbf{e}_i = E_{\text{word}}[\text{id}(t_i)] \in \mathbb{R}^{768}. \tag{2}$$

After PubMed/PMC pre-training, BioBERT places clinically related terms nearer in this space than a general English model would (Lee et al., 2020).

Position and segment.

$$E_{\text{pos}}(i) \in \mathbb{R}^{768}, \qquad E_{\text{seg}}(s) \in \mathbb{R}^{768}, \quad s = 0 \tag{3}$$

(for a single chief complaint, segment is always 0).

Full input vector (element-wise sum).

$$E(t_i)_d = E_{\text{word}}(t_i)_d + E_{\text{pos}}(i)_d + E_{\text{seg}}(0)_d, \qquad d = 1,\dots,768. \tag{4}$$

Symbol	Meaning
$E_{\text{word}}(t_i)$	Meaning of the token (dictionary lookup)
$E_{\text{pos}}(i)$	Learned vector for position i (order matters)
$E_{\text{seg}}(0)$	Segment id (here always the single-complaint segment)
$E(t_i)$	Final 768-d input for that position

Tiny numerical sketch (first three dimensions only). For token "head" at position $i = 4$, suppose (illustrative numbers):

$$\begin{aligned} E_{\text{word}} &= [0.31,\ -0.18,\ 0.52], \\ E_{\text{pos}}(4) &= [0.05,\ 0.02,\ -0.01], \\ E_{\text{seg}}(0) &= [0.00,\ 0.00,\ 0.00], \\ E(t_4) &= [0.36,\ -0.16,\ 0.51]. \end{aligned}$$

The same addition repeats for dimensions $d = 4,\dots,768$.

Stacking into $H^{(0)}$. Place each $E(t_i)$ as row i of the input matrix:

$$H^{(0)} \in \mathbb{R}^{M\times 768}. \tag{5}$$

This is the starting hidden state for encoder layer 1.

Why these dimensions?

- $M \leq 128$: sequence length (how many tokens).
- 768: hidden width of BioBERT base (how rich each token representation is).
- So $H^{(0)}$ has at most $128 \times 768 = 98{,}304$ entries per complaint—a fixed tensor shape for batched inference.

Remark 2. *The embedding table is like a dictionary: each token ID points to 768 numbers encoding meaning. Adding position tells the model where the word appears; adding segment (zero here) reserves the mechanism used in other BERT tasks with two sentences.*

What to say

"Each token becomes 768 numbers: word meaning plus position plus segment. Stack the rows into $H^{(0)}$ and feed the encoder."

4.4 Encoder layers: attention, FFN, residual, LayerNorm

Intuition

One encoder layer does two jobs: (1) let every token look at every other token (self-attention); (2) nonlinearly transform each token row alone (feed-forward network). Residuals keep old information; LayerNorm keeps numbers stable across depth.

Why this slide

The slide shows the attention formula. Examiners may ask what else sits inside each of the twelve layers—this subsection answers that.

BioBERT applies the same block $L = 12$ times. The update from layer $\ell - 1$ to layer ℓ is

$$Z^{(\ell)} = \mathrm{LayerNorm}\big(H^{(\ell-1)} + \mathrm{MultiHeadAttention}(H^{(\ell-1)})\big), \tag{6}$$

$$H^{(\ell)} = \mathrm{LayerNorm}\big(Z^{(\ell)} + \mathrm{FFN}(Z^{(\ell)})\big), \tag{7}$$

$$\mathrm{FFN}(x) = W_2\,\mathrm{ReLU}(W_1 x + b_1) + b_2. \tag{8}$$

In BERT-base / BioBERT-base: $W_1 \in \mathbb{R}^{768\times 3072}$, $W_2 \in \mathbb{R}^{3072\times 768}$, $b_1 \in \mathbb{R}^{3072}$, $b_2 \in \mathbb{R}^{768}$. The inner width $3072 = 4 \times 768$ gives the FFN capacity to transform each row nonlinearly.

Component	What it does
Multi-head attention	Twelve parallel heads; each learns different token relationships
FFN	Expand to 3072 dims with ReLU, then project back to 768
LayerNorm	Rescale activations for stable deep training
Residual (+)	Add sub-layer input to output so earlier signal is not erased

Stacking twelve times.

$$H^{(0)} \xrightarrow{\text{Layer 1}} H^{(1)} \xrightarrow{\text{Layer 2}} \cdots \xrightarrow{\text{Layer 12}} H^{(12)} = H^{(L)}.$$

Rough intuition (not a hard theorem): shallow layers emphasise words and short phrases; middle layers combine symptoms ("headache" + "feverish" + "weak"); deep layers encode broader urgency patterns. All M rows are processed in parallel at each layer.

Remark 3. *The* `[CLS]` *row of* $H^{(12)}$ *is exactly* $h_{[\mathrm{CLS}]}$ *from Equation* (1).

What to say

"Each layer: attend, add & normalise, feed-forward, add & normalise. Do that twelve times. Then read `[CLS]`."

4.5 Self-attention in detail (slide 18)

Intuition

Attention lets every token decide how much to listen to every other token. "severe" should strengthen "headache"; "no" should weaken "pain".

Why this slide

This is the heart of transformers. Know Q, K, V, the scale $\sqrt{d_k}$, and what α_j means.

Four computational steps. From the current hidden matrix H:

1. **Project** into queries, keys, and values.
2. **Score** pairwise compatibility.
3. **Normalise** scores into attention weights.
4. **Combine** value rows into context-aware outputs.

$$Q = HW_Q, \qquad K = HW_K, \qquad V = HW_V, \tag{9}$$

$$S = \frac{QK^\top}{\sqrt{d_k}}, \tag{10}$$

$$A = \mathrm{softmax}(S), \tag{11}$$

$$\text{output} = AV. \tag{12}$$

Equivalently, the compact formula on the slide (Vaswani et al., 2017):

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \qquad d_k = 64. \tag{13}$$

What Q, K, V mean in words.

- Q (query): "what am I looking for at this position?"
- K (key): "what do I offer as a matchable label?"
- V (value): "what content should be mixed if I match?"

Why divide by $\sqrt{d_k}$? Dot products of d_k-dimensional vectors grow with d_k. Without scaling, softmax saturates (weights become nearly 0 or 1), gradients vanish, and learning stalls. The factor $\sqrt{d_k}$ keeps scores in a healthy range ($d_k = 64$ per head in BioBERT base).

Attention weights α_j. For one query row, the softmax over keys assigns weight α_j to token j:

$$\alpha_j = \frac{\exp(e_j)}{\sum_{k=1}^{m} \exp(e_k)}, \qquad \sum_{j=1}^{m} \alpha_j = 1, \tag{14}$$

where e_j is the scaled score for key j and m is the sequence length (masked pads do not participate). **Meaning:** α_j is how much this position listens to token j. Symptom tokens typically receive higher weight than fillers such as "I" or "and".

Multi-head attention. Twelve heads run in parallel with different W_Q, W_K, W_V. Outputs are concatenated and projected back to 768 dimensions so heads can specialise (e.g. symptom co-occurrence vs negation vs intensity words).

Toy attention pattern (intuition only). When the query is `[CLS]`, illustrative (not measured) weights might look like:

Token	Relative weight
head / `##ache`	high
fever / `##ish`	high
weak	medium–high
I, have, a, and	low
`<PAD>`	zero (masked)

So the summary vector emphasises clinical wording, not filler words.

What to say

"$QK^\top$ scores who listens to whom; divide by $\sqrt{64}$; softmax makes weights that sum to one; multiply by V. Twelve heads, then stack twelve layers."

4.6 Softmax classification head (slide 19)

Intuition

We need five scores—one per acuity level—that behave like probabilities. Then we pick the level with the largest score. That is the entire decision rule.

Why this slide

Connects the 768-d summary to a clinical class. Say "largest probability", not bare arg max.

Linear map to five logits, then softmax:

$$\mathbf{z} = W h_{[\mathrm{CLS}]} + \mathbf{b}, \qquad W \in \mathbb{R}^{5\times 768},\ \mathbf{b} \in \mathbb{R}^5, \tag{15}$$

$$\hat{y}_c = \frac{\exp(z_c)}{\sum_{j=1}^{5} \exp(z_j)}, \qquad c = 1, \ldots, 5. \tag{16}$$

By construction $\hat{y}_c \geq 0$ and $\sum_{c=1}^{5} \hat{y}_c = 1$. $\hat{y}_c$ is the model's belief that the complaint is acuity level c.

Chosen level.

$$\hat{c} = \text{the index } c \text{ that maximises } \hat{y}_c. \tag{17}$$

Worked numerical example. Illustrative logits for levels 1–5:

$$\begin{aligned}
\mathbf{z} &= [-1.2,\ -0.5,\ 1.8,\ 0.3,\ -0.9], \\
\exp(\mathbf{z}) &\propto [0.30,\ 0.61,\ 6.05,\ 1.35,\ 0.41], \\
\sum_c \exp(z_c) &= 8.72, \\
\hat{\mathbf{y}} &\approx [0.03,\ 0.07,\ 0.69,\ 0.15,\ 0.05], \\
\hat{c} &= 3 \quad (\text{then Yellow via } f_{\text{SATS}}).
\end{aligned}$$

Deployment logs may round a dominant Yellow near 72%; the mathematics is the same: pick the largest $\hat{y}_c$.

Symbol	Meaning
$h_{[\text{CLS}]}$	768-d summary from the final encoder layer
$W, \mathbf{b}$	Learned classification weights and bias
z_c	Logit (pre-softmax score) for level c
$\hat{y}_c$	Probability of level c
$\hat{c}$	Predicted level = index of largest $\hat{y}_c$

What to say

"The head turns the summary into five probabilities that add to one. We choose the level with the highest probability."

4.7 Why Stage 3 fine-tuning (loss and one gradient step)

Intuition

PubMed pre-training teaches medical language. It does *not* teach five-class SATS acuity on short chief complaints. Stage 3 fine-tuning on the 80,000 Triagegeist pairs teaches that mapping.

Stage 1 in one sentence. Masked language modelling on PubMed/PMC teaches $P(t_i \mid \text{context})$ over biomedical text. No triage colours yet.

Stage 3 objective (what we need for the talk). Given nurse label $c^\star$ and predicted distribution $\hat{\mathbf{y}}$, minimise cross-entropy:

$$\mathcal{L} = -\sum_{c=1}^{5} \mathbf{1}[c = c^\star] \log \hat{y}_c = -\log \hat{y}_{c^\star}. \tag{18}$$

Wrong confident predictions produce large loss.

Backpropagation sketch. Gradients flow from the loss through softmax and the linear head into BioBERT:

$$\frac{\partial \mathcal{L}}{\partial W} = \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} \cdot \frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{z}} \cdot \frac{\partial \mathbf{z}}{\partial W}. \tag{19}$$

In words: (i) how wrong the probabilities are; (ii) how softmax transmits that error; (iii) how logits depend on W.

Gradient descent update.

$$W_{\text{new}} = W_{\text{old}} - \alpha \frac{\partial \mathcal{L}}{\partial W}, \tag{20}$$

with a small learning rate (thesis order $\alpha = 2 \times 10^{-5}$) so PubMed knowledge is preserved while the head and encoder adapt to urgency labels.

What to say

"We fine-tune on eighty thousand labelled complaints with cross-entropy. That is what turns BioBERT language knowledge into five acuity probabilities."

4.8 Colour map f_{SATS} and pathway $P(C)$ (slide 20)

Intuition

Training labels are integers 1–5. The hospital speaks colours, timers, and destinations. A fixed deterministic map bridges the two languages; the pathway card finishes the product.

$$f_{\text{SATS}}(\hat{c}) = \begin{cases} \text{Red} & \hat{c} = 1 \\ \text{Orange} & \hat{c} = 2 \\ \text{Yellow} & \hat{c} = 3 \\ \text{Green} & \hat{c} \in \{4, 5\} \end{cases} = C_{\text{NLP}}. \tag{21}$$

Levels 4 and 5 both map to Green (routine stream). Blue is a dignity protocol, not an NLP output class here.

Pathway card.

$$P(C) = \big(T_{\max},\ \text{destination},\ \text{actions},\ \text{escalation}\big), \tag{22}$$

aligned to KNUST University Hospital protocol:

- $T_{\max}$: maximum waiting time for that colour;
- destination: primary clinical stream;
- actions: support steps (nursing, diagnostics, HIT, ...);
- escalation: what happens if the timer expires or the patient worsens.

Protocol sketch (clinical destinations).

Colour	$T_{\max}$	Destination (summary)	Escalation
Red	0 min	Resuscitation / Surgery	Immediate senior alert
Orange	10 min	Acute medical / surgical streams	Head nurse if unclaimed
Yellow	60 min	Outpatient urgent / Pediatrics	Re-assess at timer
Green	< 4 h	Routine OPD / Dental / Eye / PPH	Return if worse

Scenario examples (tie to the gate).

Scenario	Example phrase
Gate reject	"Hello, how are you?" (no clinical content)
Red	"Crushing chest pain, cannot breathe, sweating heavily"
Orange	"Severe abdominal pain in pregnancy, bleeding"
Yellow	*"I have a headache and feel feverish. My body aches and I feel weak."*
Green	"Routine dental pain for two weeks, no fever"

What to say

"Level three becomes Yellow; Yellow becomes a card with about sixty minutes and an urgent destination. That is f_{SATS} then $P(C)$."

4.9 Clinical gate and end-to-end chain (slide 21)

Intuition

Do not waste BioBERT on "hello" or sports chat. A cheap rule-based gate asks whether the text looks clinical enough.

Why this slide

Closes the mathematical story and matches the first box on the pipeline diagram.

Relevance score. Symptom lexicon hits contribute non-negative scores $s_j(X)$. A greeting / non-clinical pattern contributes penalty $p(X)$ (prototype uses $p(X) = 0.5$ when matched).

$$g(X) = \text{clip}\Big(\sum_j s_j(X) - p(X),\ 0,\ 1\Big). \tag{23}$$

Proceed to BioBERT iff $g(X) \geq 0.35$; otherwise re-prompt for a real chief complaint and assign no colour. The threshold 0.35 rejects obvious non-medical input while accepting short valid complaints such as "chest pain".

Worked gate examples. Clinical running complaint with hits "headache" ($s_1 = 0.30$) and "feverish" ($s_2 = 0.25$), no greeting penalty:

$$\sum_j s_j = 0.55, \qquad g(X) = \text{clip}(0.55, 0, 1) = 0.55 \geq 0.35$$

$\Rightarrow$ classification proceeds.

Greeting "Hello, how are you?" with no symptoms and $p(X) = 0.5$:

$$g(X) = \text{clip}(0 - 0.5, 0, 1) = 0 < 0.35$$

$\Rightarrow$ stop; request clinical text.

Full chain (memorise).

$$X \xrightarrow{g} \tau(X) \to H^{(0)} \xrightarrow{12} h_{[\text{CLS}]} \to \hat{\mathbf{y}} \xrightarrow{f_{\text{SATS}}} C_{\text{NLP}} \to P(C_{\text{NLP}}). \tag{24}$$

If $g(X) < 0.35$, the chain stops after the gate.

1. Gate: compute $g(X)$; stop if below 0.35.
2. Tokenise and embed $\to H^{(0)}$.
3. Twelve encoder layers $\to h_{[\text{CLS}]}$.
4. Softmax head $\to \hat{\mathbf{y}}$; choose largest component $\hat{c}$.
5. Map $\hat{c} \to C_{\text{NLP}}$ and attach $P(C_{\text{NLP}})$.

What to say

"If the gate says yes, we tokenise, embed, run twelve layers, softmax, map to a colour, and emit a pathway card. That single chain is the mathematical contribution of the thesis."

5 Results, impact, and closing

5.1 Evaluation protocol (slide 22)

Why this slide

Defines what was measured before you flash numbers. Honesty about scope builds trust.

- Stratified 10% holdout from Triagegeist: $N_{\text{eval}} = 8{,}000$ (seed 42).
- Same English complaint texts and nurse acuity labels as the training distribution.
- Metrics: accuracy, macro-F1, per-class precision/recall/F1, latency (p50, p95, mean).

Remark 4 (What the numbers mean—and do not mean)**.** *Holdout scores measure agreement with historical nurse labels on **in-distribution** Triagegeist text. They are **not** prospective waiting-time or outcome trials at KNUST University Hospital (Stewart et al., 2023). Say this out loud if asked "does this prove patients get better care at KNUST?"*

What to say

"We evaluate on eight thousand held-out Triagegeist complaints. High scores mean the model matches nurse labels on that benchmark—not that we have finished a clinical trial at KNUST."

5.2 Holdout metrics and SMOTE trade-off (slide 23)

Why this slide

Shows headline performance and the safety trade-off for the rare Red class.

Metric	Baseline	SMOTE
Accuracy	0.9992	0.9985
Macro-F1	0.9977	0.9954
Recall Level 1 (Red)	0.9814	1.0000
Precision Level 1 (Red)	1.0000	0.9641
Latency p50 (ms)	100.30	95.50

How to explain the trade-off.

- **Baseline (default):** nearly perfect overall accuracy; perfect Red *precision* (when it says Red, it was Red) but Red *recall* 0.9814 (a few Reds missed).
- **SMOTE** (Chawla et al., 2002): oversamples minority classes in training. Red recall becomes 100% (safer net for critical cases) with a small drop in Red precision and a tiny drop in overall accuracy/F1.

Recall vs precision in one sentence. Recall (Red) = among true Reds, how many did we catch? Precision (Red) = among predicted Reds, how many were truly Red? For critical triage, missing a Red (low recall) is often worse than an extra Orange/Yellow alert.

What to say

"Baseline is extremely accurate with perfect Red precision. SMOTE pushes Red recall to one hundred percent for a safer clinical net, at a small precision cost."

5.3 Headline chart and confusion matrix (slides 24–25)

Why this slide

Visual confirmation of the table; confusion matrix shows where errors (if any) sit.

Bar chart. Same story as the table: baseline vs SMOTE on accuracy, macro-F1, Level 1 recall. Point at Red recall rising under SMOTE.

Confusion matrix (baseline). Rows = true labels, columns = predicted labels (or as labelled on the figure). A strong diagonal means almost all complaints are classified correctly on the holdout. If asked about off-diagonal cells: those are the rare mistakes; relate them to recall/precision.

What to say

"The confusion matrix is nearly diagonal on the holdout—errors are rare. That matches the 0.999 accuracy."

5.4 Latency and nurse vs chatbot (slide 26)

Why this slide

Answers "is this fast enough for intake?" and contrasts queue time with model time.

- Median inference ≈ 100 ms per complaint on evaluation hardware.
- Nurse path: minutes to hours to first colour (queue).
- Chatbot: English text → colour + pathway card in about a tenth of a second.
- Nurse still performs TEWS and can override (Mitchell et al., 2023).

What to say

"The model responds in about one hundred milliseconds. The nurse still owns vitals and the final clinical decision."

5.5 Anticipated intake workflow (slide 27)

Why this slide

Closes the story visually: traditional long queue vs chatbot-assisted path with TEWS in parallel.

Traditional: arrive → queue → nurse colour → often unclear destination (minutes to hours).

Chatbot-assisted: complaint → classify → pathway card (≈ 100 ms), nurse TEWS in parallel.

What to say

"We are not removing the nurse. We insert a fast text-based colour and pathway while vitals continue as usual."

5.6 Anticipated impact (slide 28)

Why this slide

Four plain claims for a mixed audience. No jargon needed here.

1. **Faster start:** suggest urgency as soon as the problem is described.
2. **Safer for critical patients:** designed to catch the most urgent cases (tie to high Red recall).
3. **Clearer next steps:** pathway card says where to go and what to do.
4. **Nurses stay in charge:** assistive tool; vitals and override remain with staff.

What to say

"Faster first signal, fewer missed critical cases on the benchmark, clearer routing, and nurses remain in control."

5.7 Selected references (slide 29)

Why this slide

You will not read this aloud. Know the big names if asked: Lee (BioBERT), Stewart (NLP triage review), SATS manual, Chawla (SMOTE), Triagegeist, plus local/context citations.

5.8 One-minute closing script

Closing

"We built an examinable pipeline from English chief complaint text to SATS colour and a KNUST pathway card, with BioBERT mathematics, a clinical gate, and strong holdout agreement on Triagegeist. The nurse keeps TEWS and override. That is the contribution of this project."

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